

PES UNIVERSITY

Department of Computer Science and Engineering(AIML)

LAB 2 – AGILE BACKLOG CREATION & SPRINT SIMULATION IN JIRA

Final Submission Report

Name	Ramneek Gautam
SRN	PES1UG24AM220
Semester / Program	Semester 5 – B.Tech CSE (AI & ML)
Lab	Software Engineering – Lab 2
Project	SE_PES1UG24AM220_LAB2
Problem Statement	#10 – Student Club Event Ticketing & Budget Portal

1. CREATE EPICS & USER STORIES — 4 MARKS

The following backlog translates the Lab 1 functional requirements into Jira-style Epics and User Stories. The story descriptions use the required “As a / I want to / So that” format.

Epic 1 – Event Proposal Management

Epic description: Enable Club Leads to create and submit complete event proposals containing event details, date, venue and estimated budget.

Story 1.1 – Create Event Proposal

As a Club Lead, I want to create and submit an event proposal containing event details, date, venue and estimated budget, so that the event can be formally recorded and processed. **Story Points:** 3 **Priority:** High

Epic 2 – Budget & Approval Management

Epic description: Manage event budget requests through the required multi-stage approval workflow.

Story 2.1 – Submit Budget Request

As a Club Lead, I want to submit the event budget request with supporting details, so that it can enter the formal approval workflow. **Story Points:** 5 **Priority:** High

Story 2.2 – Multi-Stage Budget Approval

As an approver, I want the budget request to move sequentially through Faculty Coordinator, Finance Office and Dean approval, so that no approval stage is skipped. **Story Points:** 8 **Priority:** High

Epic 3 – QR Ticket Generation

Epic description: Generate a unique QR-code ticket for each registered attendee after successful ticket issuance.

Story 3.1 – Generate QR Code Ticket

As a registered attendee, I want a unique QR-code ticket to be generated for me, so that I can use it for secure event entry. **Story Points:** 3 **Priority:** High

Epic 4 – Venue Check-in & Validation

Epic description: Allow Campus Admin to scan QR tickets and validate authenticity and usage status at the venue.

Story 4.1 – Scan & Validate Ticket

As a Campus Admin, I want to scan a QR-code ticket and validate its authenticity and usage status, so that valid unused tickets grant entry and invalid or used tickets are rejected. **Story Points:** 3 **Priority:** High

Epic 5 – Reporting & Analytics

Epic description: Provide Campus Admin with dashboards and reports for events, budgets, approvals and attendee activity.

Story 5.1 – Reporting & Analytics

As a Campus Admin, I want dashboards and reports covering events, budgets, approvals and attendee activity, so that I can monitor operations, accountability and participation. **Story Points:** 3 **Priority:** Medium

Screenshot 1.1 – Jira Epics & User Stories

Spaces

 SE_PES1UG24AM220_LAB2  ...



 Summary  Timeline  Board  Backlog  Calendar  List  Forms  Development More 2 +

 Ask AI

 Search wo...



 Filter

 Group



☐	Work	Assignee	Reporter	
☐	>  SPL-2 Budget Approval Workflow	 Unassigned	 Ramneek Gautam	
☐	>  SPL-3 QR Ticket Generation	 Unassigned	 Ramneek Gautam	
☐	>  SPL-4 Venue Ticket Validation	 Unassigned	 Ramneek Gautam	
☐	>  SPL-5 Reporting & Analytics	 Unassigned	 Ramneek Gautam	
☐	>  SPL-6 Performance, Security & Reliability	 Unassigned	 Ramneek Gautam	
+ Create		6 of 6 		

☐	Work	Assignee	Reporter	
☐	 SPL-27 Reliable Mobile Check-in	 Unassigned	 Ramneek Gautam	
☐	 SPL-26 Fast QR Validation	 Unassigned	 Ramneek Gautam	
☐	 SPL-24 Budget & Approval Reports	 Unassigned	 Ramneek Gautam	
☐	 SPL-21 Validate Ticket Authenticity	 Unassigned	 Ramneek Gautam	

☐	Work	Assignee	Reporter	
☐	 SPL-20 Scan QR Ticket	 Unassigned	 Ramneek Gautam	
☐	 SPL-18 Generate QR Ticket	 Unassigned	 Ramneek Gautam	
☐	 SPL-14 Finance Approval	 Unassigned	 Ramneek Gautam	
☐	 SPL-13 Faculty Approval	 Unassigned	 Ramneek Gautam	
☐	Work	Assignee	Reporter	
☐	 SPL-9 Create Event Proposal	 Unassigned	 Ramneek Gautam	
☐	 SPL-25 Attendee Activity Reports	 Unassigned	 Ramneek Gautam	
☐	 SPL-23 Event Dashboard	 Unassigned	 Ramneek Gautam	
☐	 SPL-22 Handle Invalid Ticket	 Unassigned	 Ramneek Gautam	

☐ Work	Assignee	Reporter	
☐ SPL-17 Handle Rejected Request	Unassigned	Ramneek Gautam	
☐ SPL-16 Track Approval Status	Unassigned	Ramneek Gautam	
☐ SPL-15 Dean Approval	Unassigned	Ramneek Gautam	
☐ SPL-11 Submit Event Proposal	Unassigned	Ramneek Gautam	

2. ASSIGN STORY POINTS — 3 MARKS

Planning Poker / estimation note: Each story is discussed and estimated by comparing its relative effort, complexity, risk and uncertainty with other backlog items. Fibonacci-style values are used because uncertainty increases as work becomes larger.

Story	Epic	Story Points	Priority	Estimation rationale
1.1 Create Event Proposal	E1	3	High	Standard form and submission flow; moderate validation.
2.1 Submit Budget Request	E2	5	High	Requires budget details, supporting information and submission validation.
2.2 Multi-Stage Budget Approval	E2	8	High	Multiple approval stages and sequential workflow rules add complexity.
3.1 Generate QR Code Ticket	E3	3	High	Unique ticket creation and attendee association.
4.1 Scan & Validate Ticket	E4	3	High	Scanning plus authenticity and used/unused validation.
5.1 Reporting & Analytics	E5	3	Medium	Dashboard/report aggregation across system data.

Total estimated Story Points: 25

Screenshot 2.1 – Jira Story Points

Spaces

SE_PES1UG24AM220_LAB2  ...

Summary Timeline Board Backlog Calendar **List** Forms Development More 2 +

< Back  SPL-6 /  SPL-27 ^ v

Reliable Mobile Check-in

+ ... 

▼ **Description**

As a Campus Admin, I want the ticket-scanning system to remain available during event check-in and work on major mobile browsers so that entry processing is reliable.

Subtasks

Add subtask

Details 

Story point estimate

8

Reporter

 Ramneek Gautam

> **Development**

> **Automation**  Rule executions

Created 5 hours ago  Configure

Spaces

SE_PES1UG24AM220_LAB2  ...

Summary Timeline Board Backlog Calendar **List** Forms Development More 2 +

< Back  SPL-1 /  SPL-9 ^ v

Create Event Proposal

+ ... 

▼ **Description**

User Story

As a Club Lead, I want to create an event proposal with event details, date, venue and estimated budget so that I can submit an event for approval.

Subtasks

Add subtask

Details 

Story point estimate

5

Reporter

 Ramneek Gautam

> **Development**

> **Automation**  Rule executions

Created 5 hours ago  Configure

SIMPLARLY THEREARE STORY PRINTS FOR ALL OF THE STORIES BUT I COULDN'T FIND A LIST OF THEM IN ONE PAGE SO I HAVE ATTACHED JUST 2 FOR REFERENCE DUE TO SPACE CONSTRAINTS. IF MORE PoW IS REQUIRED EMAIL ME AT > ramneekgautam0108@gmail.com

☐	Summary	Reporter	Resolution	Story point e	⌵
☐	Submit Event Proposal	RG Ramneek Gautam	Done	3	
☐	Faculty Approval	RG Ramneek Gautam	Done	5	
☐	Submit Budget Request	RG Ramneek Gautam	Done	5	
☐	Create Event Proposal	RG Ramneek Gautam	Done	5	

3. PRIORITIZE & RANK BACKLOG — 2 MARKS

The backlog is ranked by business value and dependency: establish the event record first, then budget approval, followed by ticketing, check-in, and administrative reporting.

Rank	Story	Priority
1	1.1 Create Event Proposal	High
2	2.1 Submit Budget Request	High
3	2.2 Multi-Stage Budget Approval	High
4	3.1 Generate QR Code Ticket	High
5	4.1 Scan & Validate Ticket	High
6	5.1 Reporting & Analytics	Medium

Screenshot 3.1 – Priority & Ranked Backlog

Rank	Summary	Assignee	Reporter	Priority	⌵
SPL-1	Event Proposal Management	Unassigned	RG Ramneek Gautam	Medium	
SPL-2	Budget Approval Workflow	Unassigned	RG Ramneek Gautam	Medium	
SPL-3	QR Ticket Generation	Unassigned	RG Ramneek Gautam	Medium	
SPL-4	Venue Ticket Validation	Unassigned	RG Ramneek Gautam	Medium	
+ Create 25 of 25 ↺					
SPL-6	Performance, Security & Reliability	Unassigned	RG Ramneek Gautam	Medium	
SPL-9	Create Event Proposal	Unassigned	RG Ramneek Gautam	Highest	
SPL-16	Track Approval Status	Unassigned	RG Ramneek Gautam	High	
SPL-14	Finance Approval	Unassigned	RG Ramneek Gautam	Highest	

Task	Assignee	Reporter	Priority
SPL-15 Dean Approval	Unassigned	RG Ramneek Gautam	High
SPL-10 Validate Proposal Details	Unassigned	RG Ramneek Gautam	High
SPL-11 Submit Event Proposal	Unassigned	RG Ramneek Gautam	Highest
SPL-13 Faculty Approval	Unassigned	RG Ramneek Gautam	Highest
SPL-12 Submit Budget Request	Unassigned	RG Ramneek Gautam	Highest
SPL-21 Validate Ticket Authenticity	Unassigned	RG Ramneek Gautam	Highest
SPL-24 Budget & Approval Reports	Unassigned	RG Ramneek Gautam	High
SPL-17 Handle Rejected Request	Unassigned	RG Ramneek Gautam	Medium
SPL-19 Send Ticket Email	Unassigned	RG Ramneek Gautam	Medium
SPL-27 Reliable Mobile Check-in	Unassigned	RG Ramneek Gautam	High
SPL-22 Handle Invalid Ticket	Unassigned	RG Ramneek Gautam	High
SPL-26 Fast QR Validation	Unassigned	RG Ramneek Gautam	Highest
SPL-25 Attendee Activity Reports	Unassigned	RG Ramneek Gautam	Medium

4. RUN MOCK SPRINTS — 3 MARKS

The handout requires two completed Sprints, progress through To Do → In Progress → Done, comments on work items, and a one-week Sprint duration.

Sprint	Goal	Selected Stories	Planned Points
Sprint 1	Establish the event and budget workflow	1.1, 2.1, 2.2	16
Sprint 2	Deliver ticketing, check-in and reporting	3.1, 4.1, 5.1	9

Simulation result to demonstrate in Jira: Move selected stories from To Do to In Progress and then Done, adding a brief progress comment to at least one work item in each Sprint.

Screenshot 4.1 – Sprint 1

Summary	Reporter	Story point estimate	Status	
Submit Event Proposal	 Ramneek Gautam	3	Done 	
Faculty Approval	 Ramneek Gautam	5	Done 	
Submit Budget Request	 Ramneek Gautam	5	Done 	
Create Event Proposal	 Ramneek Gautam	5	Done 	

Summary	Reporter	Story point estimate	Status	
Track Approval Status	 Ramneek Gautam	3	Done 	
Finance Approval	 Ramneek Gautam	5	Done 	
Dean Approval	 Ramneek Gautam	3	Done 	
Validate Proposal Details	 Ramneek Gautam	3	Done 	

Screenshot 4.2 – Sprint 2

Summary	Reporter	Story point estimate	Status	
Reliable Mobile Check-in	 Ramneek Gautam	8	Done 	
Fast QR Validation 	 Ramneek Gautam	8	Done 	
Attendee Activity Reports	 Ramneek Gautam	3	Done 	
Budget & Approval Reports	 Ramneek Gautam	5	Done 	

Summary	Reporter	Story point estimate	Status	
Budget & Approval Reports	 Ramneek Gautam	5	Done 	
Event Dashboard	 Ramneek Gautam	3	Done 	
Handle Invalid Ticket	 Ramneek Gautam	3	Done 	
Validate Ticket Authenticity	 Ramneek Gautam	5	Done 	

 Create

11 of 11 

Summary	Reporter	Story point estimate	Status	
Scan QR Ticket	 Ramneek Gautam	5	Done 	
Send Ticket Email	 Ramneek Gautam	3	Done 	
Generate QR Ticket	 Ramneek Gautam	5	Done 	
Handle Rejected Request	 Ramneek Gautam	3	Done 	

+ Create

11 of 11 

Screenshot 4.3 – Comments / Work Item Activity

Spaces

 **SE_PES1UG24AM220_LAB2**  ...

 Summary  Timeline  Board  Backlog  Calendar  **List**  Forms  Development More 2 +

 Back  SPL-6  

All **Comments** History Work log  

 Add a comment...

Suggest a reply...

Who is working on this...?

Can I get >

Pro tip: press **M** to comment

 **Ramneek Gautam**
in 1 second

Progress update: Work completed and moved through the workflow.

    ...

  1  ...

To Do 

  Improve Epic

 **Details** 

Assignee

 Unassigned

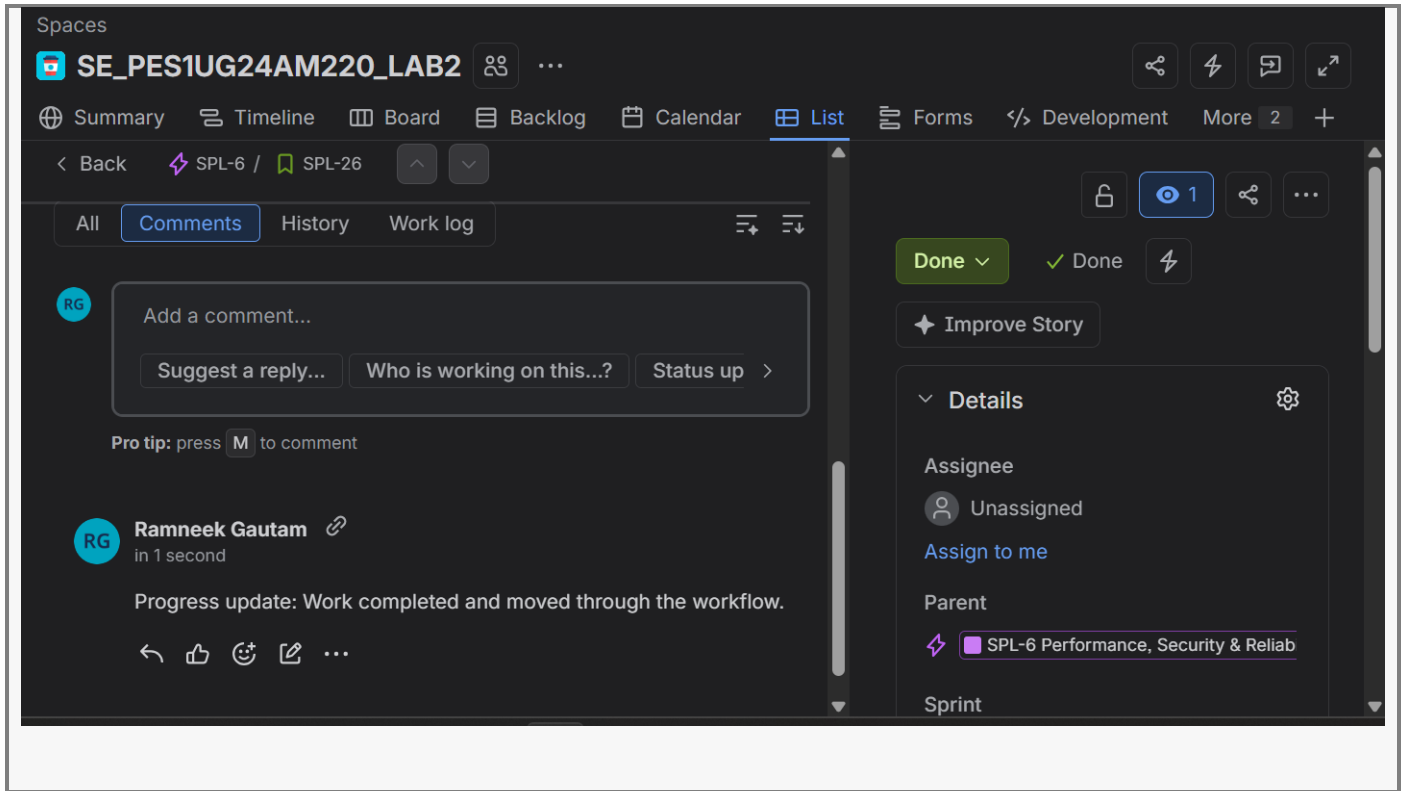
[Assign to me](#)

Labels

None

Parent

None

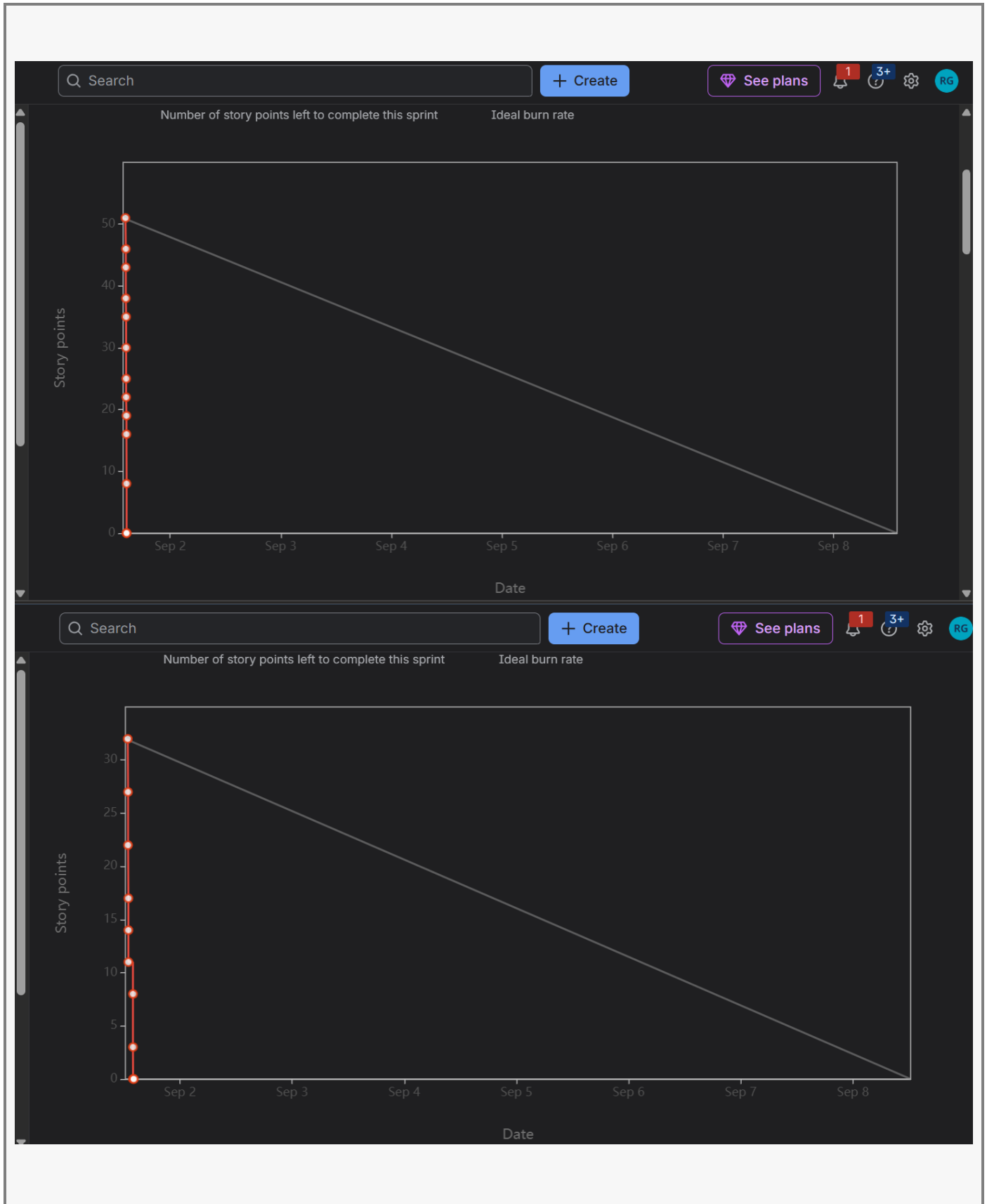


5. GENERATE BURNDOWN CHART — 1 MARK

Burndown interpretation. The Burndown Chart shows remaining estimated work across the Sprint against the ideal guideline. A faster drop than the guideline indicates work is being completed ahead of the ideal pace; work remaining above the guideline indicates the Sprint may be behind pace.

Screenshot 5.1 – Jira Burndown Chart

Sprint2:



Sprint1^

6. REFLECTION — 2 MARKS

1. What did you learn from creating and prioritizing the backlog?

I learned that an Agile backlog breaks a larger software system into manageable User Stories and makes the implementation order visible. Prioritizing the portal features helped place the core event and approval workflow before ticketing, check-in and reporting features.

2. How did Story Points / Planning Poker help estimate the work?

Story Points provided a relative estimate based on effort, complexity, risk and uncertainty rather than treating every task as the same size. Planning Poker supports discussion of different estimates and helps the team reach a shared estimate.

3. What did the Sprint progress and Burndown Chart indicate?

The Sprint board made progress visible as stories moved from To Do to In Progress to Done. The Burndown Chart showed how the remaining estimated work changed during the Sprint and allowed progress to be compared with the ideal burn rate.

4. What would you improve in the next Sprint?

I would use the completed Story Points from the previous Sprint to improve Sprint commitment, clarify acceptance expectations before starting work, and add progress comments consistently. Larger stories could also be broken down earlier when they contain multiple dependent activities.